SANTHAKUMAR RAMESH

AI/ML Engineer

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Professional Summary

AI/ML Engineer with 3 years of experience designing, building, and deploying scalable machine learning and deep learning solutions across enterprise environments. Strong expertise in Python-based ML development, data engineering, model optimization, and production deployment using cloud platforms. Proven track record of improving prediction accuracy, automating decision systems, and delivering measurable business impact through AI-driven solutions. Experienced in end-to-end ML lifecycle from data ingestion and feature engineering to model deployment, monitoring, and optimization.

Professional Experience

AI/ML Engineer (Internship) | DXC Technology | USA **July 2025 – Present**

- Designed and developed end-to-end machine learning solutions using Python, Scikit-learn, TensorFlow, and PyTorch to solve enterprise-scale prediction and classification problems, improving model accuracy by 27% while reducing manual decision-making effort across multiple business workflows.
- Built scalable data pipelines using SQL, PySpark, and cloud storage services to process structured and unstructured datasets, enabling real-time and batch inference workflows and reducing data processing latency by 35% across production ML systems.
- Implemented deep learning models including CNN and LSTM architectures for pattern recognition and time-series forecasting use cases, improving demand prediction accuracy by 22% and enabling proactive business planning.
- Performed advanced feature engineering and data preprocessing on large datasets, applying normalization, encoding, dimensionality reduction, and outlier detection techniques that improved model stability and reduced training time by 30%.
- Deployed machine learning models into production environments using Docker, Kubernetes, and CI/CD pipelines, ensuring scalable deployments, automated retraining, and consistent model performance with 99.7% system uptime.
- Integrated ML models with cloud platforms using AWS and Azure services, leveraging managed ML environments and secure storage solutions to support enterprise-grade scalability and compliance requirements.
- Developed automated model monitoring and performance tracking dashboards using Power BI and custom Python scripts, enabling early detection of model drift and improving long-term prediction reliability by 20%.
- Collaborated with data engineers, product owners, and business stakeholders to translate complex business requirements into ML-driven solutions, accelerating project delivery timelines by 25% through effective cross-functional coordination.
- Authored detailed technical documentation covering model design, data pipelines, deployment workflows, and operational guidelines, improving knowledge transfer and reducing onboarding time for new team members by 30%.

Machine Learning Engineer | Mphasis | India **July 2022 – Dec 2024**

- Developed and implemented machine learning models using Python, SQL, and Scikit-learn to automate data-driven decision processes for enterprise clients, improving operational efficiency by 24% and reducing manual analysis efforts.
- Built and maintained data preprocessing and feature engineering pipelines for large datasets, applying statistical analysis and data quality checks that improved training dataset reliability and increased model performance consistency.
- Designed predictive analytics solutions for classification and regression problems, leveraging ensemble methods and optimization techniques to improve prediction accuracy by 21% across multiple client applications.
- Worked extensively with NLP techniques including text preprocessing, vectorization, and sentiment analysis to build models that extracted insights from unstructured data, improving information retrieval accuracy by 26%.
- Utilized Apache Spark and distributed computing frameworks to process high-volume datasets efficiently, reducing batch processing time by 40% and enabling faster experimentation cycles for ML model development.
- Supported model deployment activities by containerizing ML applications using Docker and assisting in CI/CD pipeline setup, ensuring smooth transitions from development to production environments.
- Collaborated with QA, data engineering, and business teams to validate model outputs through structured testing and performance benchmarking, achieving a 97% user acceptance rate during deployment phases.
- Actively participated in Agile development processes including sprint planning, daily stand-ups, and retrospectives, contributing to on-time project delivery and continuous improvement of ML workflows.
- Created technical documentation and user guides explaining model logic, assumptions, and limitations, improving transparency and enabling business users to confidently adopt AI-driven solutions.

Technical Skills

- **Programming & Scripting:** Python, SQL, PySpark, Shell Scripting
- **Machine Learning:** Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Deep Learning:** Neural Networks, CNN, RNN, LSTM, Transformers, TensorFlow, PyTorch
- **Natural Language Processing:** Text Classification, Tokenization, Named Entity Recognition, Sentiment Analysis, Embeddings
- **Data Engineering:** Data Pipelines, ETL Processes, Data Cleaning, Data Validation, Large-Scale Data Processing
- **Big Data & Frameworks:** Apache Spark, Hadoop Ecosystem, Distributed Computing
- **Cloud Platforms:** AWS (S3, EC2, SageMaker), Azure ML, Cloud-based Model Deployment
- **MLOps & Deployment:** Docker, Kubernetes, CI/CD Pipelines, Model Versioning, Monitoring
- **Databases & Storage:** SQL Databases, NoSQL, Data Warehouses
- **Visualization & Reporting:** Power BI, Tableau, Model Performance Dashboards
- **Tools & Collaboration:** Git, Jira, Confluence, Agile Scrum, Cross-functional Collaboration.

Education

Master's in data science and application | University at Buffalo, New York, USA **Jan 2025 – Dec 2025**

Bachelors in Electronics and Communication | Rajalakshmi Engineering college, Chennai, India **Oct 2020 – May 2024**